

STELLA (YUXUAN) JIANG

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EDUCATION

University of California, Irvine

Expected Jun. 2027

B.S. in Computer Science | GPA: 3.88/4.0 | Dean's Honor List

Coursework: Data Structures and Algorithms, Machine Learning, Database Management, Software Engineering

EXPERIENCE

AI Algorithm Intern | Alibaba Group

Jul. 2026 - Present

Hangzhou, China | Tmall Campus digital services and retail ecosystem: 800+ universities and nearly 20 million students.

- Built a Python/FastAPI RAG service for campus knowledge search and benchmarked BM25, BGE, and RRF on 778 entries and 6,562 queries; BGE reached **76.40% Top-1** and **91.34% Top-3** and became the production retriever.
- Protected the campus assistant from answering beyond its evidence by adding a **Coverage Gate** and deterministic/LLM risk guards with shadow/enforce modes, causing unsupported, privacy-sensitive, and safety-sensitive requests to fail closed before generation.
- Improved campus product discovery with five-source retrieval combining ItemCF, field-aware BM25, dense embeddings, constraints, deduplication, and Weighted RRF; in a 1,000-request offline ablation, **Recall@10 rate rose from 70% to 85%** when the relevant item entered the candidate pool.
- Built structured intent/profile services and batch pipelines for personalization, transforming 100M raw events into **86.3M training records for 987,992 users** while keeping offline training outside latency-sensitive request paths.
- Delivered four community content workflows with replaceable model providers, Pydantic contracts, fact guards, and atomic quota counters that preserved quota on invalid outputs; verified them with **371 backend/smoke tests**, 9 browser tests, and regression over 999 valid posts.

Software Engineer | FessorAI, UCI Capstone Collaboration

Apr. 2026 - Jun. 2026

Irvine, CA | Full-stack AI education platform for structured textbook generation and in-page editing.

- Built a **six-stage LLM workflow with deterministic orchestration** that turns an outline into textbook content, interactive exercises, code, images, and video, using shared adapters to run OpenAI, Gemini, Claude, and Grok through consistent generation contracts.
- Separated long-running generation from web requests with **Django, Celery, Redis, and PostgreSQL**, adding dependency tracking, durable job state, retries, failure recovery, and synchronized progress updates for the React frontend.
- Enabled in-place textbook revision by extending a **28-tool Page Assistant** with native function calling and conversation memory; streamed text and tool events through Django Channels, Redis, and WebSockets into navigable React blocks for human review.
- Fixed long-form editing failures caused by a 1,000-character truncation by sending **full-section context** to tool calls, allowing the assistant to revise lengthy textbook sections without dropping surrounding material.

Founder & Software Engineer | FocusTrail

Jan. 2026 - Present

Irvine, CA | UCI Startup Fund-supported local-first AI execution coach for task planning and recovery; selected for Alibaba Tmall Campus OPC (5/50).

- Turned unstructured planning inputs into actionable work by building a **Gemma-powered multimodal pipeline** that converts PDFs, screenshots, and notes into structured task DAGs with explicit dependency edges for planning, progress tracking, and recovery.
- Made live plans resilient to changing priorities with a versioned replanning runtime supporting dependency-aware update/defer/insert operations, completed-work protection, plan validation, state reconciliation, deterministic fallback, and undo, so schedule changes could be applied without corrupting completed work or dependent tasks.
- Shipped **local-first React/Node.js/Express/SSE** web and macOS/Windows prototypes with rule/LLM activity classification, local/cloud model routing, and **pre-inference privacy filters** that block sensitive-app content before any model call.

AI Engineer Intern | NextTier

Dec. 2025 - Feb. 2026

Sacramento, CA | Policy-grounded receipt and expense-report audit copilot for structured extraction, policy validation, and approve/review/reject decisions.

- Built an **AI expense-audit copilot** that analyzed receipts and expense reports, validated transaction details, and returned approve/review/reject recommendations with policy-grounded explanations, reducing median review time from **6.4 to 2.1 minutes per report**.
- Developed a **LangChain RAG pipeline** over company expense policies using document chunking, OpenAI embeddings, a vector database, semantic retrieval, and context-grounded generation.
- Implemented structured outputs and **LLM function calling** to query internal expense APIs for missing fields, duplicates, and prior records; improved record-backed answer accuracy from **82.4% to 91.1%**, reduced model calls from **2.5 to 1.6 per audit**, and cut token cost by **36%**.
- Designed a modular **FastAPI/Pydantic** backend that separated stateless audit orchestration and validation from stateful policy, receipt, and audit-record storage, improving scalability, maintainability, and reliability; containerized the backend with Docker.
- Created an **MLflow evaluation pipeline** over synthetic expense reports. Ran **36 MLflow experiments** across chunk size, overlap, top-k, prompt, and model configurations; selected a configuration that improved accuracy from **84.6% to 90.9%**, reduced p95 latency from **5.9s to 3.6s (39%)**.

Software Engineer Intern | Beijing Jiuqi Software Co., Ltd.

Jun. 2025 - Sep. 2025

Beijing, China | Enterprise AI-agent platform and data services for government and financial-reporting workflows.

- Migrated an internal enterprise AI-agent platform from **Dify to Nuwa GPT** to support natural-language access to government and financial-reporting data while preserving legacy workflow behavior across production use cases.
- Turned heterogeneous enterprise data into reusable agent capabilities by developing **Python/Flask REST services** on Alibaba Cloud ECS/OSS that normalized structured and unstructured sources into agent-callable tools.
- Standardized Docker/Kubernetes CI/CD for 10+ distributed services, **cutting deployment time by 40%** and eliminating manual rollback steps; added Function Compute health checks and runtime metrics to surface post-migration failures earlier.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL

AI & ML: RAG, Information Retrieval, Ranking, LLM Agents, Function Calling, Structured Outputs, Model Evaluation, LangChain, MLflow, Local LLMs

Frontend: React, WebSocket, Server-Sent Events

Backend: FastAPI, Django, Flask, Node.js/Express, Celery, REST APIs

Databases: PostgreSQL, Redis, SQLite, SQLAlchemy, Vector Databases

DevOps: Docker, Kubernetes, CI/CD, GitHub Actions, Alibaba Cloud